Assignment 9

Electricity and Electronics EBN 111

6 June 2022

Test Information			
Maximum marks:	20	Full marks:	20
Due Date:	6 June 11:59	Open/Closed book:	Open
Total number of pages (this page included)		8	

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Assistant Lecturers: D. Wepener, K. Mphaga, N. Matic, J. Thiselton, L. Zheve

Important

1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers, including complex numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By uploading your assessment, you declare the following:**

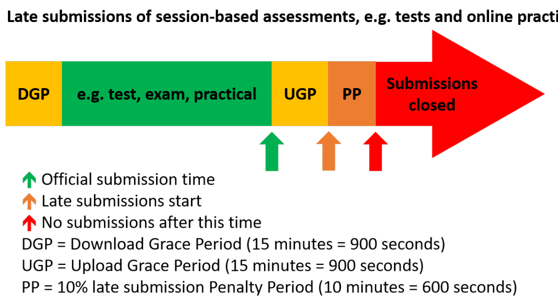
   The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment.

   I hereby declare that the work herewith is completely my own and that no parts have been copied in any way from current or previous students or other sources.

Online assessment submission rules and time-line

The following rules and time-line apply for all tests for EBN111/122 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the test for students to download all documents related to the test and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the test.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the test.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2021.ams@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers on the AMS during this period without incurring any penalties.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheet during this period, the student incurs a 10% penalty to their final test mark.
8. No late submissions are accepted after the PP.

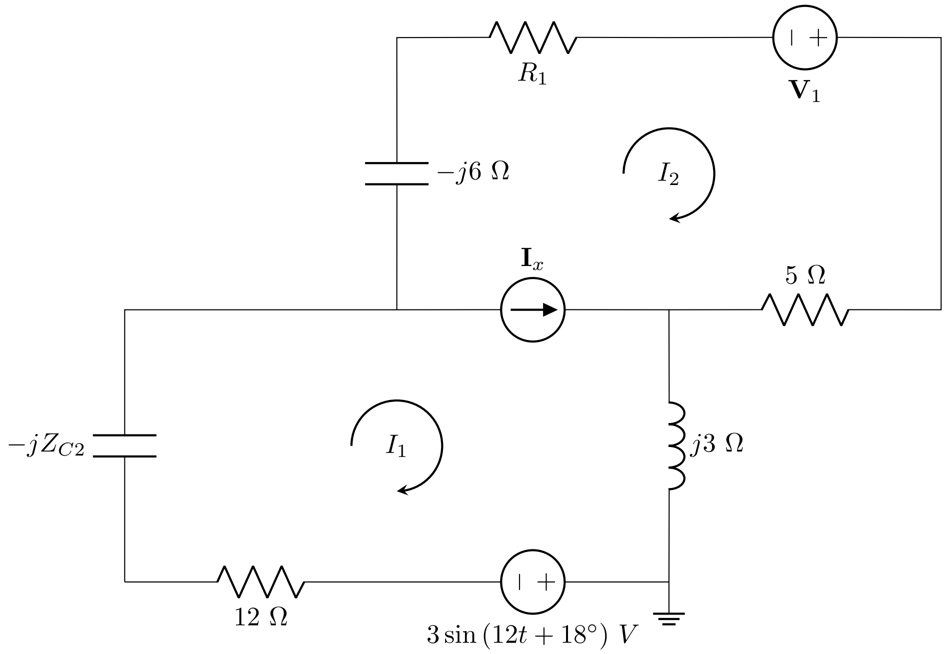


Additional guide for Complex numbers

The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

- **Correct**: *1-j2* or *-3+j4*
  - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
  - Capital *J* is however incorrect, and *1-2j* is also not acceptable.
  - If a complex number has no real part, for example *0+j2*, write *0+j2* as the final answer.
- **Correct**:
  - With angle in degrees: 12&*ang*;34&*deg*;
  - With angle in radians: 12&*ang*;0.5934
  - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect**:
  - *1-i2* (Don't use *i*, use *j*)
  - *1 - j2* (Don't use spaces between characters)
  - 12 &*ang*; 34 &*deg*; (Don't use spaces between characters)
  - *j10+1* (Don't write the imaginary part before the real part for rectangular form.)
  - *-3+4j* (Don't write the *j* after the number for imaginary numbers, write *j*4 instead of 4*j*)

Section 1

Questions 1 to 2 [4]	
Given: $R_1,\ Z_{C2},\ \mathbf{V}_1$.	
Use mesh analysis to set up an equation of the following form: $$\alpha\mathbf{I}_1 + \beta\mathbf{I}_2 = -3$$ 	
01	Determine α [2]
02	Determine β [2]

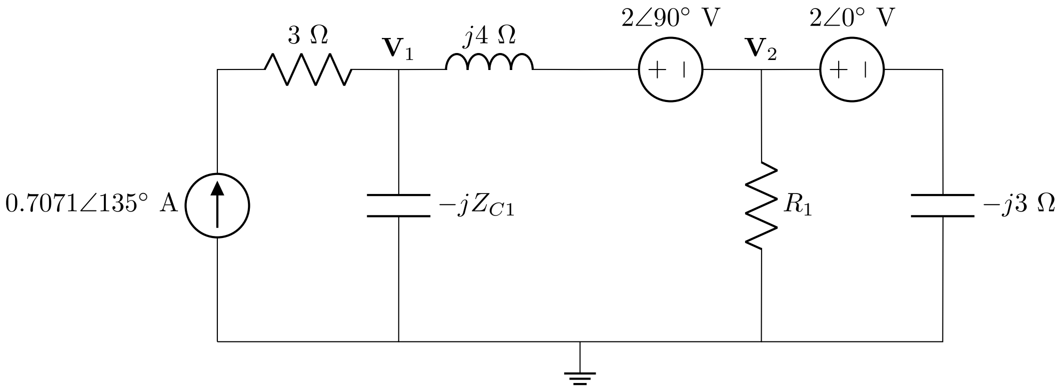
Questions 3 to 6 [4]	
Given: $Z_{C1},\ R_1$.	

Use nodal analysis to set up two equations of the following form:

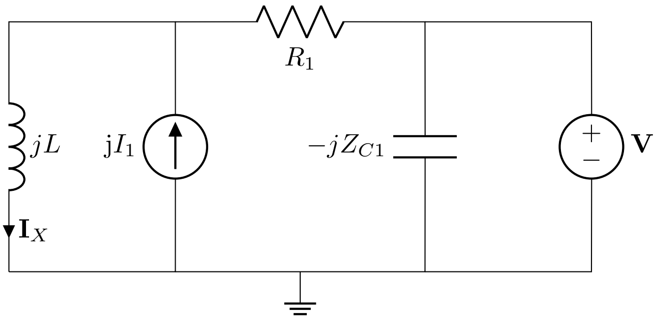
$$a\mathbf{V}_1 + b\mathbf{V}_2 = 2$$

$$c\mathbf{V}_1 + d\mathbf{V}_2 = 8 + j6$$

Determine the phasor coefficients a, b, c, and d in **rectangular form**



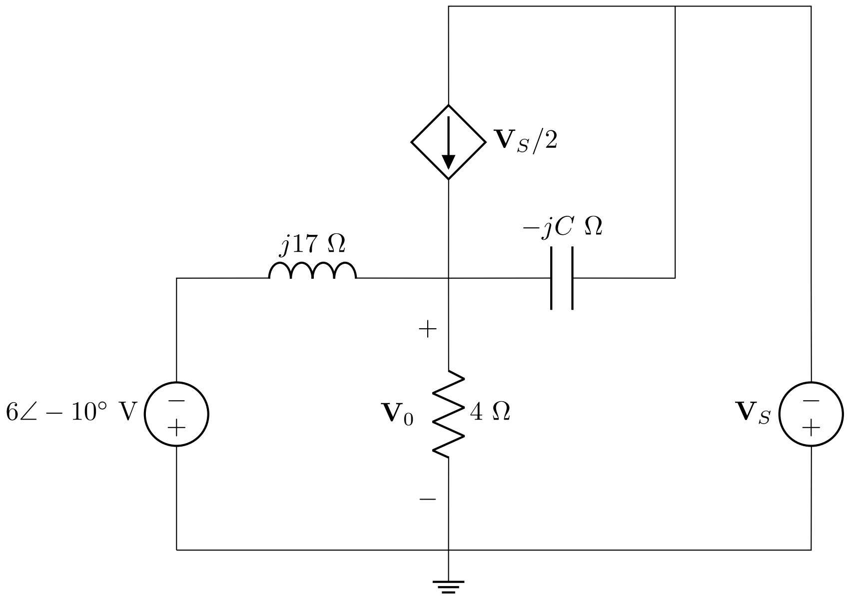
03	a [1]
04	b [1]
05	c [1]
06	d [1]

Question 7 [2]	
Given: $L,\ R_1,\ Z_{C1},\ \mathbf{V},\ I_1$.	
	
07	Use superposition to determine $\mathbf{I}_X^A$, the independent contribution of the current source to $\mathbf{I}_X$ in **rectangular form** [2]

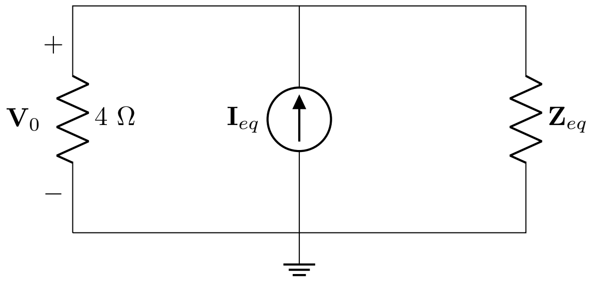
Question 8 [2]

Given: $\mathbf{V}_S,\ C.$

Circuit:A

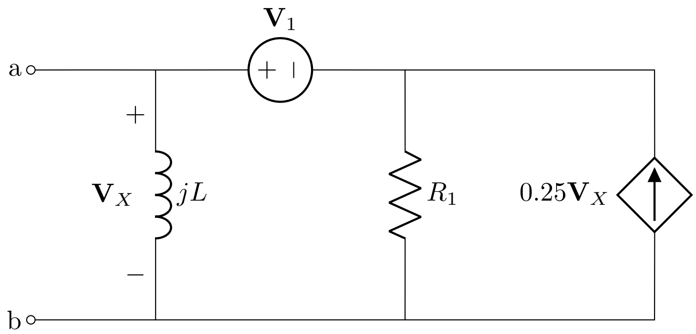


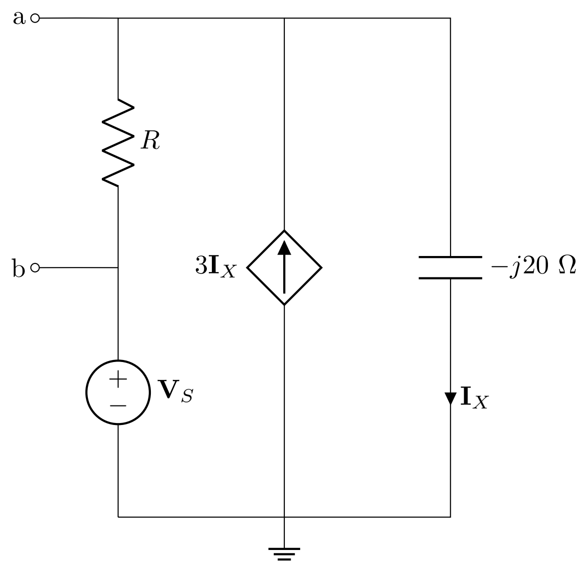
Circuit:B



08	Use source transformation to obtain $\mathbf{I}_{eq}$ in Circuit B such that it is equivalent to Circuit A. Give your answer in **polar form** as $\mathbf{I}_{eq} = r\angle\theta^\circ$, where $r \geq 0$ and $-180^\circ \leq \theta \leq 180^\circ$. [2]

Questions 9 to 10 [4]	
Given: $R_1,\ L,\ \mathbf{V}_1$.	
Find the Norton equivalent current $\mathbf{I}_N$ and impedance $\mathbf{Z}_N$ with respect to terminals *a*-*b* in **rectangular form**.	
09	$\mathbf{I}_N$ [2]
10	$\mathbf{Z}_N$ [2]



Questions 11 to 12 [4]	
Given: $\mathbf{V}_S,\ R$.	
Determine the Thevenin equivalent voltage $\mathbf{V}_{Th}$ and impedance $\mathbf{Z}_{Th}$ with respect to terminals a-b in **polar form** as $\mathbf{V}_{Th} = r\angle\theta°$, where $r \geq 0$ and $-180° \leq \theta \leq 180°$. The same format applies to $\mathbf{Z}_{Th}$ 	
11	$\mathbf{V}_{Th}$ [2]
12	$\mathbf{Z}_{Th}$ [2]

Total Section 1: [20]

Grand Total : [20]